

# AMEER SALMAN M

## Forward Deployed Engineer — Data & AI Platforms

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### SUMMARY

Data engineer with **8+ years** designing, building and optimising batch and real-time pipelines, lakehouse platforms and automated ETL workflows at petabyte scale. Currently a Forward Deployed Engineer at Databricks, delivering **\$180K+/year in verified cloud savings**, contributing an algebraic migration-validation engine into the open-source **Lakebridge** tool, and building runtime governance for AI agents. Previously seven years at Amazon Pay, cutting reporting SLAs **99% (48 hours to 15 minutes)**, processing **400M+ transactions monthly**, and leading the RBI-mandated regulatory migration of the payments data estate. Deep expertise across **Apache Spark internals, Unity Catalog, Delta Lake and the AWS big-data ecosystem**, combined with customer-facing field engineering.

### TECHNICAL SKILLS

|                  |                                                                                                                          |               |                                                                                                                |
|------------------|--------------------------------------------------------------------------------------------------------------------------|---------------|----------------------------------------------------------------------------------------------------------------|
| LANGUAGES        | Python, SQL, Scala, Java, Bash                                                                                           | PLATFORMS     | Databricks (Unity Catalog, Delta Lake, Apps, Asset Bundles, System Tables), AWS, Azure                         |
| DISTRIBUTED      | Apache Spark (internals, Spark SQL, PySpark), EMR, Photon                                                                | AWS           | Glue, S3, Athena, Redshift, Lambda, MWA, Kinesis, DynamoDB, EC2, CloudWatch, IAM, QuickSight                   |
| DATA ENGINEERING | ETL/ELT, real-time & batch streaming, warehousing, dimensional modelling, reconciliation, schema evolution, partitioning | MIGRATION     | Lakehouse architecture, JDBC pushdown, algebraic data fingerprinting, Redshift / Teradata / Oracle / Snowflake |
| FINOPS           | DBU cost profiling, execution-plan optimisation, Photon tuning, partition pruning                                        | AI GOVERNANCE | Runtime policy enforcement, OAuth / OBO token brokering, SQL AST parsing, tamper-evident audit, EU AI Act      |
| DEVOPS / IAC     | AWS CDK, Terraform, Docker, Git, CI/CD pipelines, automated packaging                                                    | TESTING       | Pytest, ScalaTest, Testcontainers, GitHub Actions, Maven multi-profile builds                                  |

### EXPERIENCE

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| Forward Deployed Engineer · Databricks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | JUL 2024 – PRESENT |
| REMOTE & CUSTOMER ONSITE   DATA & AI PLATFORMS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                    |
| <ul style="list-style-type: none"><li><b>Workload cost optimisation:</b> profiled and re-engineered high-DBU production Spark and Delta Lake workloads for enterprise accounts, generating <b>\$180,000+/year in verified savings</b> including a <b>50% compute reduction</b> on a single critical workload, via partition pruning, shuffle minimisation and join restructuring.</li><li><b>Algebraic migration validation:</b> engineered a data-fingerprinting framework inside Databricks' open-source <b>Lakebridge</b> migration tool, eliminating the need to move raw tables across the network to prove post-migration parity.</li><li><b>Pushdown computation:</b> implemented JDBC pushdown aggregation (sums, power sums, dual-channel collision hashes) on source databases, cutting egress <b>~48,000x</b> (gigabytes to ~50 KB) and accelerating validation <b>8.1x</b> on multi-million to billion-row datasets.</li><li><b>Exact-difference solver:</b> authored a recovery solver with <b>zero false positives/negatives across 15.8B+ rows</b>, enabling full verification across Redshift, Teradata, Oracle and Snowflake; fixed 3 serialisation and timestamp-drift bugs safeguarding fidelity across all field migrations.</li><li><b>Production-scale optimisation sandbox:</b> built a simulation engine on <b>Delta shallow clones</b> and <b>Unity Catalog lineage</b> to test Photon, runtime upgrades and query rewrites against production-volume data without risking live pipelines or duplicating storage.</li><li><b>Automated parity verification:</b> implemented bidirectional <b>EXCEPT ALL</b> multiset checking proving <b>100% byte-level parity across 4.45B+ records</b> while demonstrating <b>10.19x</b> end-to-end workload acceleration; fail-closed identity boundaries and deferred DBU cost attribution across <b>30+ pipeline families</b>.</li></ul> |                    |

- **Agentic AI runtime gateway:** architected a multi-workspace security gateway as a Databricks App intercepting, risk-scoring, auditing and remediating autonomous AI-agent operations before production execution, via a **6-checkpoint pipeline** at **sub-30ms** per-checkpoint latency.
- **Zero-trust identity & forensics:** engineered OAuth on-behalf-of token brokering with zero credential sharing plus a hash-chained append-only Unity Catalog audit ledger, delivering compliance evidence for **EU AI Act Article 14** and SOC 2; automated reversal of unauthorised mutations through Delta time travel.
- **Spark connector modernisation:** reverse-engineered Microsoft's archived Spark SQL Server connector for **Databricks Runtime 14.3–18.x** (Spark 3.5 & 4.0), unblocking enterprise Unity Catalog migrations while preserving **~15× bulk write throughput** over generic JDBC; productised with a public-API compatibility layer and automated regression gates, adopted across multiple engagements.

Data Engineer · Amazon Pay / Amazon Payments

OCT 2017 — JUL 2024

BENGALURU & CHENNAI | DATA ENGINEER II ON EXIT

- **Near real-time pipeline architecture:** designed and deployed an end-to-end event-driven pipeline on **S3, Lambda, Glue and Athena**, cutting business reporting SLAs **99% — from 48 hours to 15 minutes** for critical payment data streams.
- **Large-scale ingestion & reconciliation:** engineered multi-source pipelines integrating **15+ upstream services** across **120+ ETL jobs**, processing **400M+ transactions/month** with real-time financial categorisation and reconciliation.
- **Big-data optimisation:** automated high-throughput ETL across multi-terabyte scale on **EMR (Spark), EC2 and S3**, reducing manual operational workload **60% monthly**.
- **Orchestration & elastic compute:** implemented enterprise workflow orchestration on **AWS MWAA** and **AWS CDK**, dynamically provisioning and tearing down transient **EMR clusters** for cost-efficient ETL execution.
- **Infrastructure as Code & CI/CD:** spearheaded IaC adoption using **AWS CDK** across multiple microservices and pipelines; built an internal CI/CD framework for **AWS Glue** packaging dependencies as egg/zip binaries with **Pytest** validation and staged artifact publishing.
- **RBI regulatory migration:** led end-to-end migration of **Redshift clusters, QuickSight dashboards and S3 data lakes** to the **AWS Mumbai region** for Reserve Bank of India payment aggregator and gateway mandates — multi-terabyte cutovers with zero downtime and verified reconciliation.
- **Audit remediation:** conducted deep-dive root-cause analysis on reconciliation discrepancies with banking partners, resolving audit exceptions valued at **\$20+ billion** across India regulatory accounts.
- **Monitoring & alerting:** configured custom **CloudWatch** metrics, automated log parsers and multi-tier alarms for real-time discrepancy detection — zero data loss at **99.9% pipeline uptime**.
- **Operational automation:** built ingestion and validation tooling in **Python and Selenium** for high-volume operational workflows, saving **15+ FTEs** of manual effort monthly; authored SOPs and design documents enabling partner-team self-service.

SELECTED FRAMEWORKS & CONTRIBUTIONS

- **DataPrint (Databricks Lakebridge)** — algebraic migration verifier. 8.1× faster sign-offs, ~48,000× less network egress, validated to 15.8B rows with zero false positives.
- **ShadowClone** — production-risk-free performance sandbox. Enabled confident Photon and runtime upgrades across 30+ pipeline families, demonstrating up to 10.19× execution gains.
- **AgentGuard / LakeGuard** — autonomous AI security gateway. Multi-workspace runtime enforcement with SQL AST firewall, token broker and automated Delta rollback; generates EU AI Act and SOC 2 audit trails.
- **Spark MSSQL Connector** — maintained field distribution. Public-API compatibility layer across Scala 2.12/2.13 and Spark 3.5/4.0, preserving 15× bulk ingestion performance.

EDUCATION

DEGREE

B.E. Mechanical Engineering — Park College of Engineering & Technology, Coimbatore, 2017